

Bird Strike Analysis

This presentation analyzes bird strike data from the FAA between 2000–2011, examining the impact on aviation safety and exploring potential mitigation strategies.



The Problem of Bird Strikes

Bird strikes pose a significant threat to aircraft safety, potentially causing damage to structures, engines, and flight control systems. They are most common during takeoff, landing, and low-altitude flight.

Safety Hazard

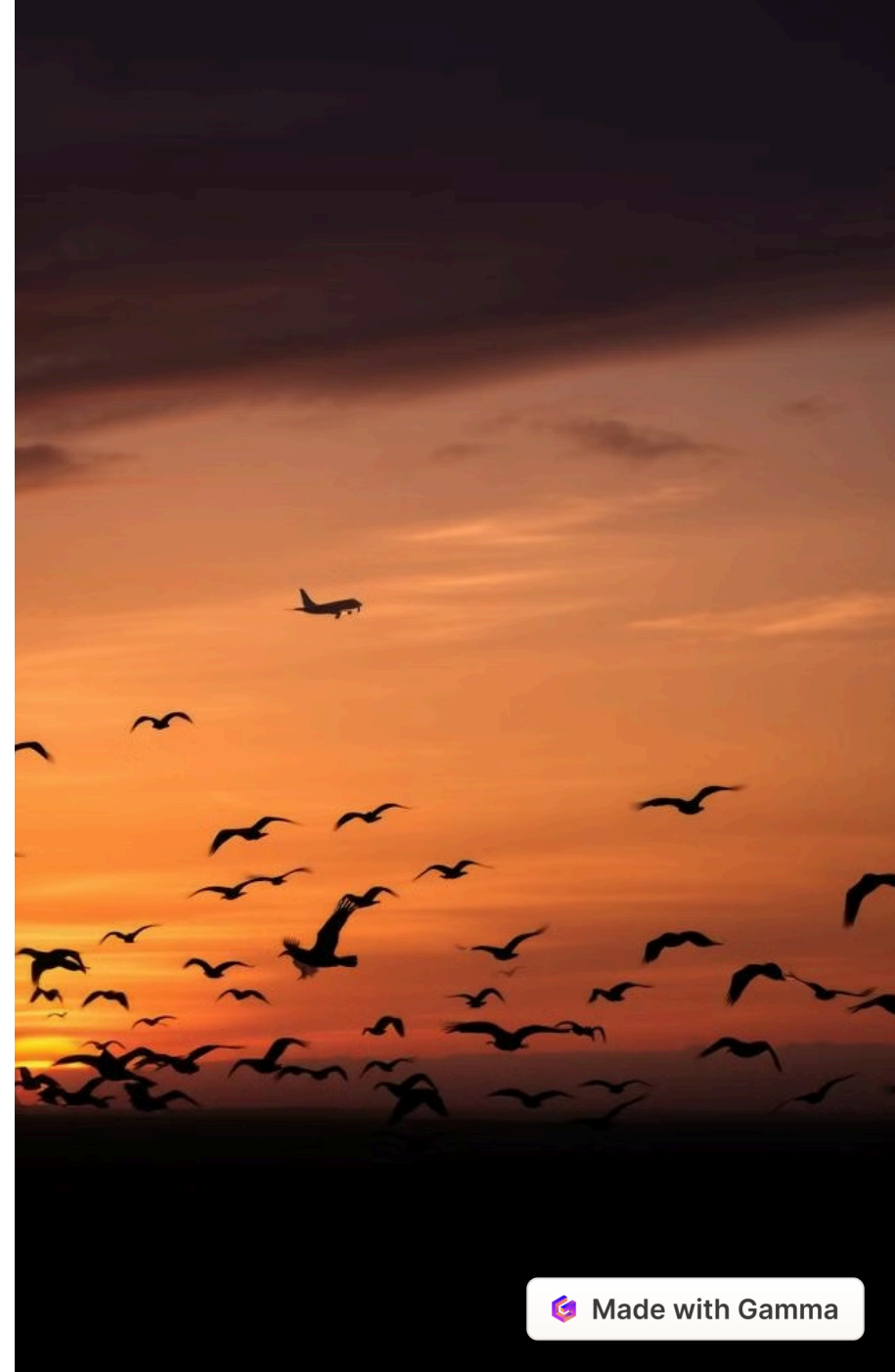
Collisions can lead to severe damage, engine failure, and even catastrophic accidents.

Common Occurrences

Most frequent at altitudes below 3,000 feet, near airports during ascent or descent.

Costly Consequences

Result in expensive repairs, flight delays, and potential loss of life.



Data Collection and Cleaning

The dataset, obtained from a unified source, contains 25,429 records of bird strikes from January 2, 2000, to December 31, 2011. Data cleaning involved handling missing values, correcting data types, and creating new columns for analysis.

1 Data Source

Unified mentor dataset with 25,429 records.

2 Data Cleaning

Addressed missing values, corrected data types, and created new columns.

3 Data Enrichment

Created columns for year, month, day, season, and region.



Key Insights from the Data

The average height of bird strikes is 799 feet, suggesting they occur during takeoff or landing. The maximum number of birds in a single strike was 942, with an average of 2.7 birds per strike.

Strike Altitude

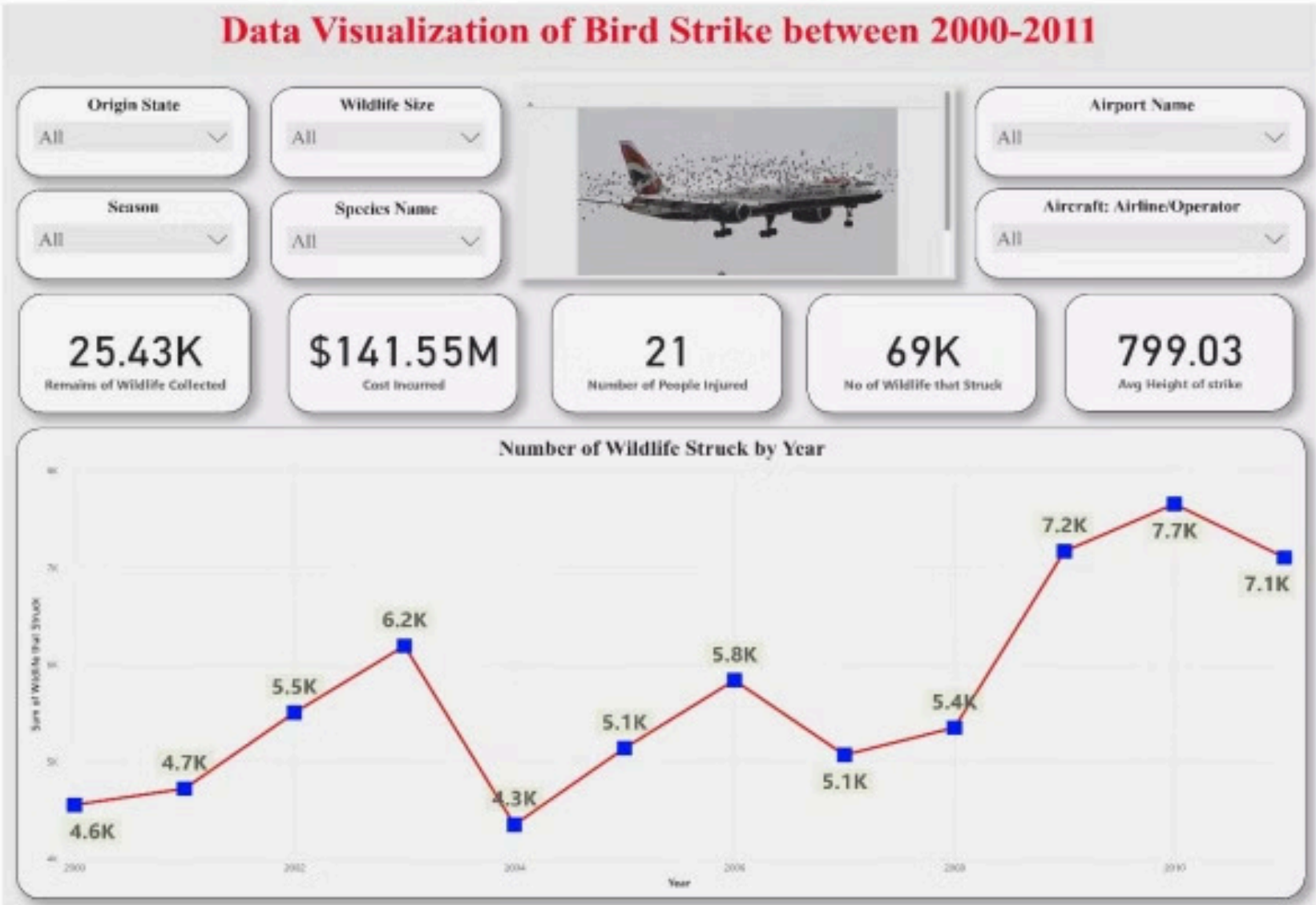
Average height: 799 feet, indicating takeoff or landing phases.

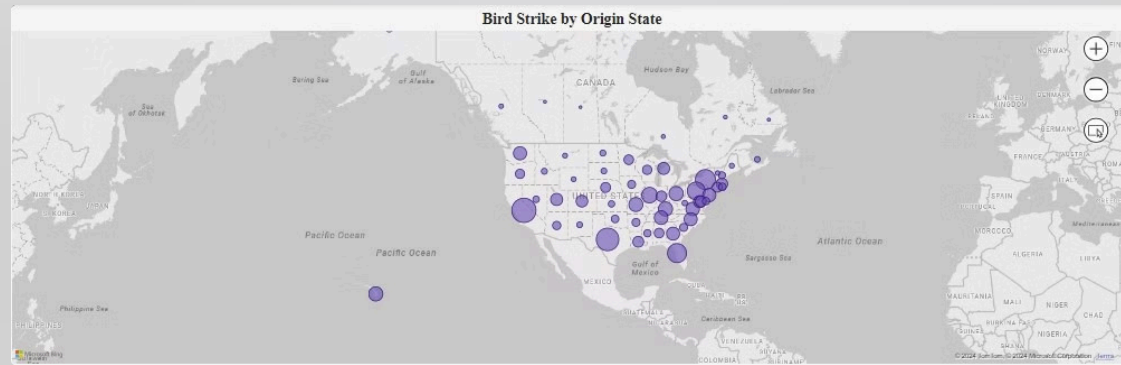
Strike Magnitude

Maximum birds in a strike: 942, average: 2.7 birds.

Cost of Strikes

Cost range: \$0 – \$12,397,750, with a mean of \$5,566 per strike.





Temporal and Spatial Patterns

2010 recorded the highest number of bird strikes. California had the most strikes among states, and the summer season saw the highest frequency.

1

2010

Year with the highest recorded
bird strikes.

2

California

State with the most bird strike
incidents.

3

Summer

Season with the highest
frequency of bird strikes.

Impact and Species Involved

Most strikes didn't significantly impact flights, but some led to precautionary landings or aborted takeoffs. Canada geese and white-tailed deer caused notable damage.



Minimal Impact

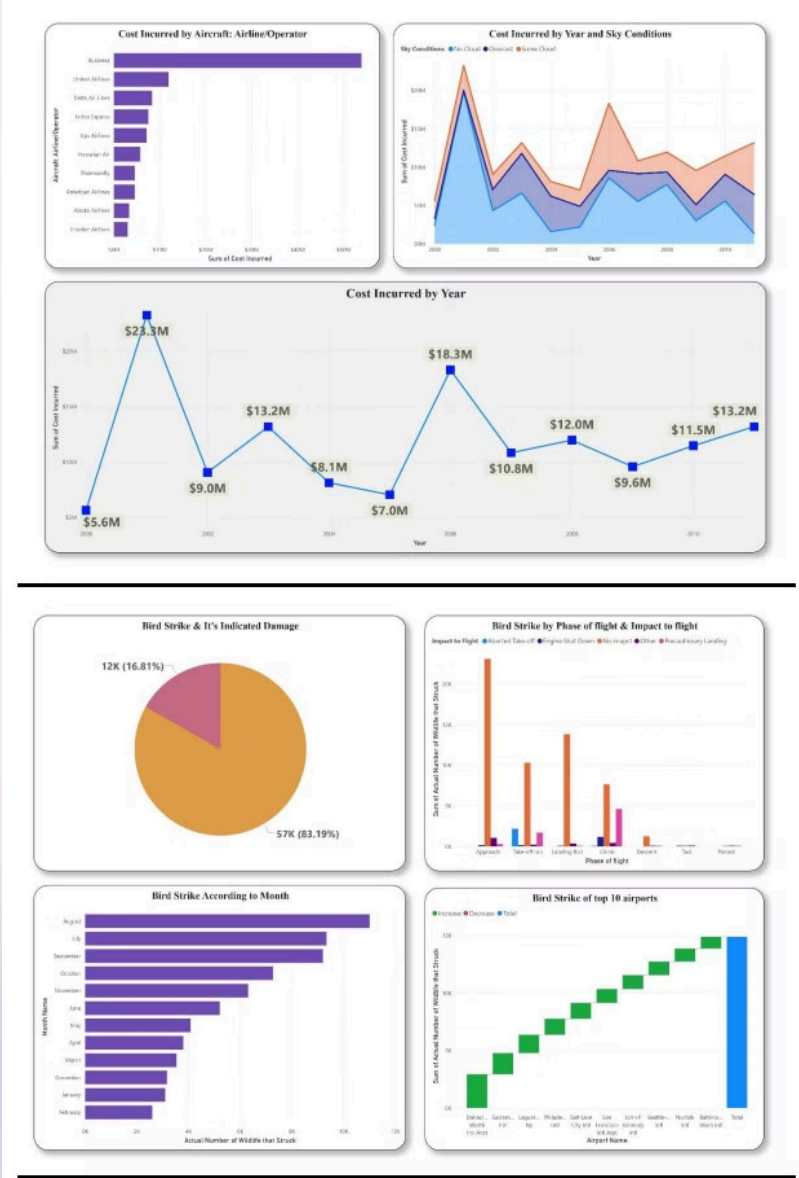
Most strikes had little effect on flights.

Precautionary Actions

Some strikes led to precautionary landings or aborted takeoffs.

Notable Damage

Canada geese and white-tailed deer caused significant damage.



Factors Influencing Bird Strikes

Most strikes occurred during no precipitation, with rain, fog, and snow following. Dallas/Fort Worth International Airport recorded the highest number of strikes.



Weather

Most strikes happened in clear conditions, followed by rain, fog, and snow.



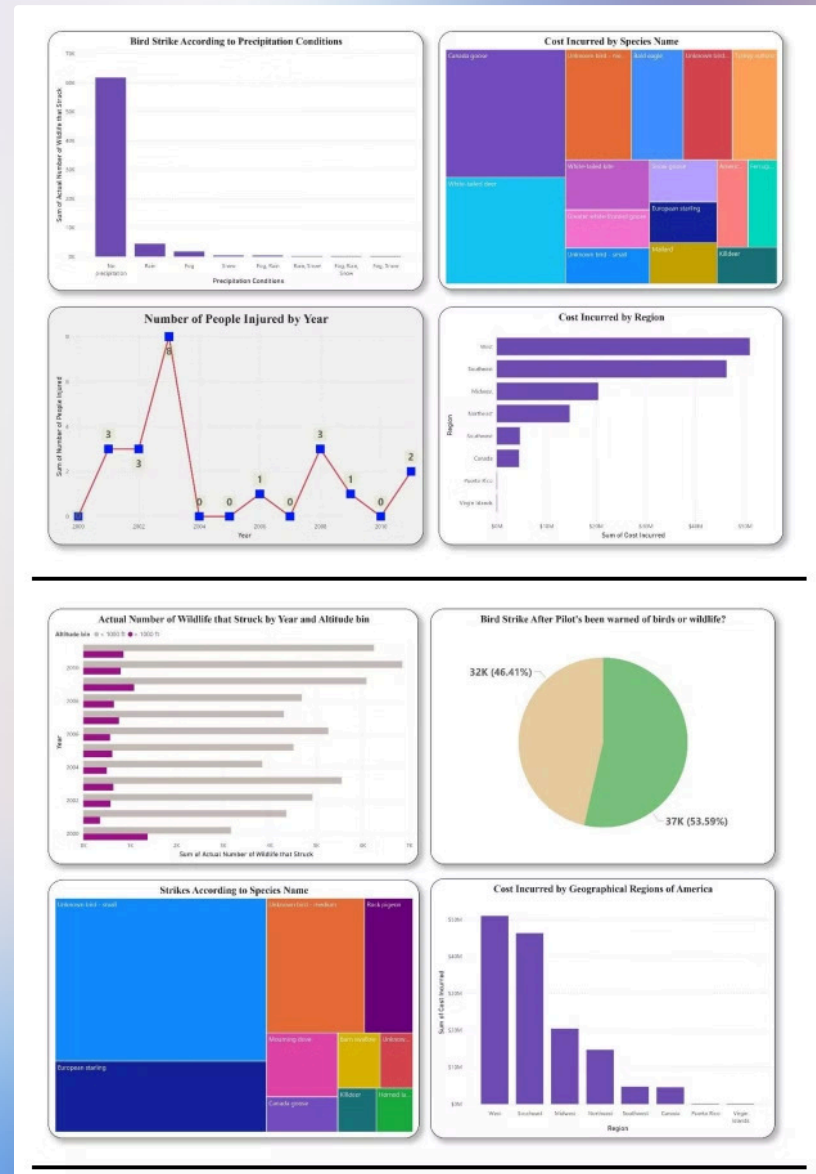
Airport

Dallas/Fort Worth International Airport had the most recorded strikes.



Region

Western states like California, Washington, and Arizona experienced the most strikes.



Mitigation and Prevention

Strategies include habitat management, wildlife detection systems, aircraft design improvements, pilot training, and bird deterrent technologies.

1 Habitat Management

Modify airport environments to reduce attractive food and water sources for birds.

2 Detection Systems

Use radar and other technologies to track bird activity and warn pilots.

3 Aircraft Design

Strengthen windshields and engines to better withstand bird impacts.

